FINALTERM EXAMINATION
EDU431
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Q.1:

Define the term measurement?

Answer:

Measurement:

Measurement is the process by which the attributes or dimensions of some object (both physical and abstract) are quantified.

Q.2 :

What is the table of specifications?

Answer:

One of the tools used by the teachers to develop a blueprint for the test is called the table specification.

Q.3:

Write any three limitations of MCQS. (From Topic 75)

Answer:

Limitations of MCQS:

1. Take a long time to construct in order to avoid arbitrary and ambiguous questions.
2. Also require careful preparation to avoid preponderance of questions testimony recall.
3. Provide cues that do not exist in practice.

Q.4:

What is the predictive validity?

Answer:

Predictive Validity:

Predictive validity refers that how much test can predict future performance with reference to some defined standard. **Q.5:**

What is the diagnostic assessment? (From Topic 4) Diagnostic assessment:

Diagnostic assessment is the type of the assessment that determines the causes intellectual, emotional, and environmental of persistent learning difficulties.

Q.6:

Describe the Bloom's taxonomy of cognitive domain? Answer:

Cognitive Domain:

There are six levels of the cognitive domain:

1. Knowledge
2. Comprehension
3. Application
4. Analysis
5. Synthesis
6. Evaluation

Q.7:

What is the test security? (From Topic 106) Answer:

Test Security:

Test security is the process intended or reveal flaws in the security mechanism of an information system that protect data and maintain functionality as intended.

Q.8:

How to design rubric?(From Topic 112) Answer:

We can design a rubric by the following some steps:

1. Identify a learning goal.
2. Choose outcomes that may be measured.
3. Develop or adapt an existing rubric.
4. Share it with students.

Q.9:

How to assemble or packaging a test?(From Topic 103) Answer:

Assembling Test (Packaging the Test):

1. Grouping together items of similar format
2. Arranging test items from easy to hard
3. Properly spacing items
4. Keeping items and options on the same page
5. Placing illustrations near the descriptions
6. Checking the answer key
7. Deciding how students will record their answers
8. Providing space for test taker's name and date
9. Checking test directions for clarity
10. Proofreading the test **Q.10:**

Write a brief note on the restricted response essay type items?

Answer:

Pose a specific problem for which student needs to recall suitable information, organize it, derive a defensible conclusion, and express it within the given limits the questions (like 400 word, 30 min. etc.).

Example:

List the similarities and differences in the process of cell division in meiosis and mitosis.

A variety of learning outcomes (Linn, 2000) can be checked by using this format of **essay question. Some of which** are;

1. Analysis of *relationship.*
2. *Compare and contrast* positions.
3. Explain *cause-effect* relationship.
4. *Organize data* and support a viewpoint.
5. *Formulate hypotheses*.

6. Point out *strengths and weaknesses.*
7. *Integrate data* from various resources

Teacher can use this type of questions under the following conditions when:

1. *Supplying information is required* instead of simple recognition.
2. *Limited numbers of content areas are needed to be tested*.

If it is feared that students may help each other unfairly during the test which is difficult to control. (Especially if the number of students is large and class teacher cannot invigilate all of them as keenly as needed). Student is to decide length, content to be included, its organization and order of presentation min the answer.

Q.11:

How to write essay type questions? (From Topic 90) Answer:

Following guidelines are provided by Haldane and Downing (1989) for writing essay type items when developing a test.

1. Frame questions so that the examinee's task is explicitly defined.
2. Specify the value and an approximate time limit for each question.
3. Employ a larger number of questions that require relatively short answers rather than only a few questions that require long answers.
4. Do not employ optional questions.
5. Verify a question's quality by writing a trial response to the question.
6. Prepare a tentative scoring key in advance of considering examinee responses.
7. Score all answers to one question before scoring the next question.
8. Make prior decisions regarding treatment of such factors as spelling and punctuation.
9. Evaluate essay responses anonymously.

10. **Q.12:**

Explain Criterion and predictive Validity.

Answer:

Criterion validity refers how much test can measure some performance with reference to some defined standard.

For example:

The student is expected to be able to speak fluently after completing a course. So, the test is valid if it can measure spoken ability.

Predictive validity refers that how much test can predict future performance with reference to some defined standard.

Q.1:

Explain the Taxonomy levels of Cognitive.

Answer:

Bloom's Taxonomy of Cognitive Development

Bloom identified six levels within the cognitive domain, from the simple recall or recognition of facts, as the lowest level, through increasingly more complex and abstract mental levels, to the highest order which is classified as evaluation. A description of the six levels as well as verb examples that represent intellectual activity are listed here.

Knowledge is defined as remembering of previously learned material. This may involve the recall of a wide range of material, from specific facts to complete theories, but all that is required is the bringing to mind of the appropriate information. Knowledge represents the lowest level of learning outcomes in the cognitive domain.

Verbs: arrange, define, duplicate, label, list, memorize, name, order, recognize, relate, recall, repeat, and reproduce state.

Comprehension is defined as the ability to grasp the meaning of material.
This may be shown by translating material from one form to another (words to numbers), by interpreting material (explaining or summarizing), and by estimating furture trends (predicting consequences or effects). These learning outcomes go one step beyond the simple remembering of material, and represent the lowest level of understanding.

Verbs: classify, describe, discuss, explain, express, identify, indicate, locate, recognize, report, restate, review, select, and translate.

Application refers to the ability to use learned material in new and concrete situations. This may include the application of such things as rules, methods, concepts, principles, laws, and theories. Learning outcomes in this area require a higher level of understanding than those under comprehension.

Verbs: apply, choose, demonstrate, dramatize, employ, illustrate, interpret, operate, practice, schedule, sketch, solve, use, write.

Analysis refers to the ability to break down material into its component parts so that its organizational structure may be understood. This may include the identification of the parts, analysis of the relationships between parts, and recognition of the organizational principles involved. Learning outcomes here represent a higher intellectual level than comprehension and application because they require an understanding of both the content and the structural form of the material.

Verbs: analyze, appraise, calculate, categorize, compare, contrast, criticize, differentiate, discriminate, distinguish, examine, experiment, question, and test.

Synthesis refers to the ability to put parts together to form a new whole. This may involve the production of a unique communication (theme or speech), a plan of operations (research proposal), or a set of abstract relations (scheme for classifying information). Learning outcomes in this area stress creative behaviors, with major emphasis on the formulation of new patterns or structures.

Verbs: arrange, assemble, collect, compose, construct, create, design, develop, formulate, manage, organize, plan, prepare, propose, set up, write.

Evaluation is concerned with the ability to judge the value of material (statement, novel, poem, research report) for a given purpose. The judgements are to be based on definite criteria. These may be internal criteria (organization) or external criteria (relevance to the purpose) and the student may determine the criteria or be given them. Learning outcomes in this area are highest in the cognitive hierarchy because they contain elements of all the other categories, plus conscious value judgements based on clearly defined criteria.

Verbs: appraise, argue, assess, attach, choose compare, defend estimate, judge, predict, rate, core, select, support, value, evaluate.

Q.2:

What is the formula for the find out the item discrimination?(From Topic 97)

Answer:

ID is = (Upper Group % Correct) – (Lower Group % Correct)

Q.3:

What are the guidelines for the essay type questions? (Repeat)

Answer:

Following guidelines are provided by Haldane and Downing (1989) for writing essay type items when developing a test.

10. Frame questions so that the examinee's task is explicitly defined.
11. Specify the value and an approximate time limit for each question.
12. Employ a larger number of questions that require relatively short answers rather than only a few questions that require long answers.
13. Do not employ optional questions.
14. Verify a question's quality by writing a trial response to the question.
15. Prepare a tentative scoring key in advance of considering examinee responses.
16. Score all answers to one question before scoring the next question.
17. Make prior decisions regarding treatment of such factors as spelling and punctuation.

18. Evaluate essay responses anonymously.

Q.4:

Describe the advantages for the essay writing test?

Answer:

1. Effective for assessing higher order abilities: *analyses synthesize and evaluate*.
2. It is comparatively *less time consuming* to develop such items.
3. Emphasizes essential *communicational skills*.
4. *Guessing* is eliminated

Q.5:

Explain Item difficulty in IRT.

Answer:

Item difficulty in IRT:

The application of Item difficulty in IRT is defined as the ability at which the probability of success on the item is .**5 on a logit scale, which is also known as threshold difficulty** (see Fig. 3). An item's level of difficulty is another factor affecting an individual's probability of responding in a particular way. An item that has a high level of difficulty will be less likely to be answered correctly by an examinee with low ability than an item that has a low level of difficulty (i.e., an easy item).



Figure 3: Three item characteristic curves with the same discrimination but different levels of difficulty (Baker, 2001, p. 10)

In Figure 2, three item characteristic curves are presented on the same graph. All have the same level of discrimination but differ with respect to difficulty. The left- hand curve represents an easy item because the probability of correct response is high for low-ability examinees and approaches 1 for high-ability examinees. The center curve represents an item of medium difficulty because the probability of correct response is low at the lowest ability levels, around 0.5 in the middle of the ability scale and near 1 at the highest ability levels. The right-hand curve represents a hard item. The probability of correct response is low for most of the ability scale and increases only when the higher ability levels are reached. Even at the highest ability level shown (+3), the probability of correct response is only 0.8 for the most difficult item.

Q.6:

What is the Rubric, Write the element of the rubric?

Answer:

Rubric:

A rubric is an explicit set of criteria used for assessing a particular type of work or performance and provides more details than a single grade or mark. It is a set of scoring guidelines for evaluating student work.

Scores or levels identified in a scoring rubric must be descriptive, not merely judgmental in nature. **Example:** Define the level of rubric as “Writing is clear and thoughts are complete” as compared to “Excellent.”

Elements of Rubric:

A rubric includes:

1. Score
2. Criteria
3. Levels of performance
4. Descriptors

Q.7:

What is the self-assessment?

Answer:

Self-assessment:

Self-assessment is a process of formative assessment during which students reflect on and evaluate the quality of their work and their learning, judge the degree to which they reflect explicitly stated goals or criteria, identify strengths and weaknesses in their work, and revise accordingly.

Q.8:

What is Analytic rubric?

Answer:

Each criterion is assessed separately by using different descriptive ratings.

Each criterion is given a separate score.

Final score is made up of adding each component parts.

It takes more time to score but gives detailed more feedback.

Q.9:

Describe the Advantages of Self-Assessment.

Answer:

Advantages:

- Self-assessment can promote learner responsibility and independence.
- Self-assessment tasks encourage student ownership of the learning.
- Self-assessment tasks shift the focus from something imposed by someone else to a potential partnership.
- Self-assessment emphasizes the formative aspects of assessment.
- Self-assessment encourages a focus on process.
- Self-assessment can accommodate diversity of learners‟ readiness, experience and backgrounds.

- **Q.10:**

Why use peer assessment?

Answer:

- Falchikov (2007) reminds us that peer learning builds on a process that is part of our development from the earliest years of life (it is the practice of formal education and the centrality of the teacher that makes us lose sight of this).
- Peer feedback can encourage collaborative learning through interchange about what constitutes good work.
- If the course wants to promote peer learning and collaboration in other ways, then the assessment tasks need to align with this. It is also important to recognize the extra work that peer learning activities may require from students through the assessment. Boud,

- Cohen & Sampson (1999) observe that “if students are expected to put more effort into a course through their engagement in peer learning activities, then it may be necessary to have this effort recognized through a commensurate shift in assessment focus” (p.416).

- Peer learning draws on the “cognitive apprenticeship model” (Kvale, 2006).
- Students can help each other to make sense of the gaps in their learning and understanding and to get a more sophisticated grasp of the learning process.
- The conversation around the assessment process is enhanced. Research evidence indicates that peer feedback can be used very effectively in the development of students‟ writing skills.
- Students engaged in commentary on the work of others can heighten their own capacity for judgment and making intellectual choices.
- Students receiving feedback from their peers can get a wider range of ideas about their work to promote development and improvement.
- Peer evaluation helps to lessen the power imbalance between teachers and students and can enhance the students‟ status in the learning process.
- The focus of peer feedback can be on process, encouraging students to clarify, review and edit their ideas.
- It is possible to give immediate feedback, so formative learning can be enhanced. Peer assessment processes can help students learn how to receive and give feedback which is an important part of most work contexts.
- Peer assessment aligns with the notion that an important part of the learning process is gradually understanding and articulating the values and standards of a “community of practice” (Wenger, 1999, cited in Falchikov, 2007, p.129). Drawing on Wenger‟s ideas,

- Falchikov suggests that “learning involves active participation in a „community of practice“ in which members of the community determine and structure their own practices, and construct identities in relation to these communities” (2007, p.129). Peer commentary in the assessment process initiates into the community to hear, experiment with and gradually internalize the norms of the community

Q.11:

What is the Syllabus criteria for the higher education in Pakistan?

Answer:

Q.12:

What are the objective test item types?

Answer:

There are three types of items which are included in the category of selection type items or the objective test items types

1. : multiple choice items,
2. alternative form or
3. True-false and matching.

Q.13:

Define Reliability with examples.

Answer:

Reliability refers to the extent to which assessment results are consistent. The assessment that maintains the consistency in results is known as reliable.

"Reliability is the extent to which a measurement instrument or procedure yields the same results as on repeated trails (Carmines and Zeller, 1979)"

For example, a calculator cannot be reliable if it gives wrong calculations or results. The weight of an object is not reliable if machine is not working properly. Same is the case of the results obtained from assessments.

Q14:

What is meant by the term of unreliable results?

Answer:

The unreliable results of assessment mean inconsistency. Keeping these views in mind, we can trust the results if these are consistent. It is the evidence of an accurate system of assessment. Keeping in view the significance of reliability, different methods are used to attain reliability. For example, test retest procedure, the alternative–test form procedure and the split halves procedure.

Q.1:

Define the evaluation?

Answer:

Evaluation is process of making a value judgment against intended learning outcomes

And behavior, to decide quality and extent of learning.

Q.2:

What is the item difficulty?

Answer:

Item difficulty is a measure of the proportion of examinees who responded to an item correctly.

Q.3:

What are the importance of the student self-assessment?

Answer:

1. **Clarifying, sharing, and understanding learning intentions and criteria for success**– getting the students to really understand what their classroom experience will be and how their success will be measured.
2. **Engineering effective classroom discussions, activities, and learning tasks that elicit evidence of learning**– developing effective classroom instructional strategies that allow for the measurement of success.
3. **Providing feedback that moves learning forward**– working with students to provide them the information they need to better understand problems and solutions.
4. **Activating learners as instructional resources for one another**– getting students involved with each other in discussions and working groups can help improve student learning.
5. **Activating learners as owners of their learning**– self-regulation of learning leads to student performance improvement

Q.4:

What is the alternative form of test?

Answer:

Alternative form of question requires students to select any one of the two given categories. The categories may be True-false, yes-no, correct-incorrect or

fact-opinion (Airasian, 1994). Alternative form of items is most suitable for measuring lower level learning outcomes.

Q.5:

What is the maximum and typical performance?

Answer:

Maximum Performance:

What a person can do. For example how you drive on a drivers test. Happens when someone knows they are being observed.

Typical Performance:

What a person *will* do – ie. How you drive normally. To observe typical performance, you have to do it when people don't know they are being observed or be there on enough occasions that people get used to the examinee, or don't know what they are being tested on.

Q.6:

What is the restricted response item?

Answer:

Pose a specific problem for which student needs to recall suitable information, organize it, derive a defensible conclusion, and express it within the given limits the questions (like 400 word, 30 min. etc.).

Q.7:

What is face validity and content validity?

Answer:

Face Validity:

- Does the test appear to test what it is supposed to be tested?
- A drawing test must be related to figures, so the question paper should appear with figures. A test that asks for filling some blanks must have blanks on paper. A multiple choice question must have options along with it.

Content Validity:

Content validity refers to the contents a test is suppose to measure.

For instance, a book has ten chapters the chapters are the contents of the book.

Q.8:

Describe the curriculum in Pakistan at higher level education?

Answer:

- In 1973's constitution, Government placed education on the concurrent list.
- Federal Government took the responsibility to determine the curriculum text boards, policy, planning and standards of education.
- In December, 1976 Federal Government appointed University Grant commission (now HEC) as the competent authority to look after the curriculum revision work at bachelor level and onwards

- It also look after the degrees, certificates and diplomas awarded by degree colleges, universities and other institutions of HEC.

Q.9:

Describe the item difficulty in CTT?

Answer:

At the item level, the Classical Test Theory (CTT) model is relatively easy. CTT does not raise a complex theoretical model to relate an examinee's ability to the probability of success on a particular item. Instead, CTT collectively considers a pool of examinees and empirically examines their success rate on an item (assuming it is dichotomously scored). This success rate of a particular pool of examinees on an item, well known as the p value of the item, is used as the index for the item difficulty (Linden& Hambleton, 2004). (Actually, it is an inverse indicator of item difficulty, with higher value indicating an easier item). In CTT, the item difficulty index p (p value), is the proportion of examinees correct on an item, expresses item difficulty on an Item. Item difficulty in CTT is simply calculated by the percentage of students that correctly answered the item as refers to the p value which range from .00 to 1.00. The values closer to 1 more easy will be the item and conversely the values near to .00 the more difficult will be the item. The values lie somewhere in the middle i.e. 0.4 to 0.6 will refer to moderate item difficulty index.

Item difficulty: the percentage of students that correctly answered the item. The range is from 0% to 100%, or more typically written as a proportion of 0.0 to 1.00. The higher the value, the easier will be the item.

Calculation: Divide the number of students who got an item correct by the total number of students who answered it.

Ideal value: Slightly higher than midway between chance (1.00 divided by the number of choices) and a perfect score (1.00) for the item. For example, on a four- alternative, multiplechoice item, the random guessing level is 1.00/4 = 0.25; therefore, the optimal difficulty level is .25 + (1.00 - .25) / 2 = 0.62. On a true-false question, the guessing level is (1.00/2 = .50) and, therefore, the optimal difficulty level is .50+(1.00- .50)/2 = 0.75. P-values above 0.90 are very easy items and should be carefully reviewed based on the instructor's purpose. For example, if the instructor is using easy "warm-up" questions or aiming for student mastery, than some items with p values above .90 may be warranted. In contrast, if an instructor is mainly interested in differences among students, these items may not be worth testing. P-values below 0.20 are very difficult items and should be reviewed for possible confusing language, removed from subsequent exams, and/or identified as an area for re-instruction. If almost all of the students get the item wrong, there is either a problem with the item or students were not able to learn the concept. However, if an instructor is trying to determine the top percentage of students that learned a certain concept, this highly difficult item may be necessary. The optimal item difficulty depends on the question-type and on the number of possible distractors. Many test experts believe that for a maximum discrimination between high and low achievers, the optimal levels (adjusting for guessing) are:

Q.10:

Explain the character tics of the Curve.

Answer:

The item characteristic curve is considered as the basic building block of item response theory; all the other constructs of the theory depend upon this curve. Therefore, significant attention is devoted to this curve and its role within the theory. There are two methodological properties of an item characteristic curve that are used to describe it: one is the difficulty which under item response theory describes the item functions along the ability scale. For example an easy item functions among the low- ability examinees and a hard item functions among the highability examinees; thus, difficulty is a location index. The second technical property is discrimination, which describes how well an item can differentiate between examinees having abilities below the item location and those having abilities above the item location (Baker, 2001).

An item characteristic curve is the graphical representation of the probability of answering an item correctly with the level of ability on the construct being measured. It gives you a picture of:

a. The item difficulty
b. Discrimination power
c. The probability of answering correctly by guessing

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Q.11:

What are the steps for packing test material?

Answer:

- Grouping together items of similar format
- Arranging test items from easy to hard
- Properly spacing items
- Keeping items and options on the same page
- Position illustration near description (if diagram given)

- Randomness of answer key
- Determine how students record answers (separate sheet or same sheet)
- Providing space for test taker‘s details
- Proofread the test (typographical and grammatical error)
- Test directions

Q.12:

What is the Prohibited Professional Test Security Practices? Answer:

1. Give students access to test content or secure test questions before test administration.
2. Copy reproduces or uses in any manner inconsistent with test security measures of secure assessment material.
3. Review actual test items before, during or after the test administration.
4. Fail to follow directions for the distribution and return of secure material or fail to account for any secure materials before, during or after test administration.
5. Leave a testing room unsupervised at any time.
6. Make test answers available to students or permit the use of any supplemental or reference materials during test.
7. Assist a student directly or indirectly (gestures, pointing, prompting etc.,)
8. Disclose or discuss the content of the tests with students, teachers, parents, other educators or community members before, during or after testing.
9. Making any changes to student’s responses.

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The End …..

Best of Luck

Remember me in your prayers..

By :

Nadia Khan ..!

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EDU 431 final paper

Evaluation

Evaluation is process of making a value judgment against intended learning outcomes and behavior, to decide quality and extent of learning. Evaluation is always related to your purpose, you aligned your purpose of teaching with what students achieved at the end, with their quality and quantity of learning

High stake testing

The use of test and assessment to make decisions that are of prominent educational, financial, or social impact.

- Decision based on high stake testing
- Promotion to next grade
- Awarding of diploma or degree
- Evaluation of school performance
- Incentives and accountability of school staff

Holistic

Holistic rubric is probably more appropriate for extended response essay type items involving a student‘s abilities to synthesize and create and when no definite answer can be prespecified.

Bloom taxonomy

Bloom’s Taxonomy of Learning Objectives Bloom‘s Taxonomy was presented by Benjamin Bloom in the 1956, consists of a framework with most common objectives of classroom instruction.

Bloom’s Taxonomy of Learning Objectives

Those dealing in three different domains and further sub categories in these domains.

1. Cognitive
2. Affective
3. Psychomotor

Cognitive Domain:

- Knowledge
- Comprehension
- Application
- Analysis
- Synthesis
- Evaluation

Affective Domain

- Receiving

- Responding
- Valuing
- Organization
- Characterization

Psychomotor Domain

- Perception
- Set
- Guided Response
- Mechanism
- Complex covert Response
- Adaption
- Origination

Example of knowledge comprehension and application level

Knowledge:

It is defined as the remembering of previously learned material. This may involve the recall of a wide range of facts, procedures principals and generals, the recall of procedures and the processes.

Sample Question: Define the 6 levels of Bloom's taxonomy of the cognitive domain.

Comprehension:

It is defined as the ability to grasp the meaning of the material. Individual can make use of the content or idea being communicated without necessarily related it to other content and seeing its fullest implications.

Sample Question: explain the purpose of Bloom's taxonomy of the cognitive domain.

Application:

It refers to the ability to use the previously learned material in new and concrete situations. The abstractions may be in the shape of universal ideas, rules of methods.

Sample Question: write an instructional objective for each level of Bloom's taxonomy.

Item characteristic curve

To analyze items using IRT, the main thing need to consider is item characteristic curve (ICC). The item characteristic curve is considered as the basic building block of item response theory.

It gives a picture of:

- The item difficulty
- Discrimination power
- The probability of answering correctly by guessing

Advantages of anecdotal record:

Following are the advantages of anecdotal records.

- It depicts actual behaviors in natural situations.
- Facilitate gathering evidence on the events that are exceptional but significant.
- Beneficial for students with less communication skills.

Steps to design rubric

Basic Steps to design Rubric

- Identify a learning goal.
- Choose outcomes that may be measured.
- Develop or adapt existing rubric.
- Share it with students

Test security

Educational decisions

In the classroom, decisions are constantly being made. Teachers face huge numbers of dilemmas every day. These decisions can be of following nature

Types of Educational Decisions

- Instructional
- Grading
- Diagnostic
- Selecting
- Placement
- Counseling and Guidance
- Program or Curriculum
- Administrative

These types of decisions are taken at different levels. Some are decided at Board/Administrative level while some are taken at school management level and other are taken in classrooms by teachers

Alternatives

The list of suggested solutions which may include words, numbers, symbols or phrases are called alternatives (also called choices or options).

The correct alternative in each item is called the answer while all incorrect or less appropriate alternatives are called distractors or foiled and the student's task is to select the correct or best alternative from all the options.

Formative assessment

Formative assessment provides feedback and information during the instructional process, while learning is taking place, and while learning is occurring. Formative assessment measures student progress but it can also assess your own progress as an instructor.

Elements of rubric

A rubric includes:

- Score
- Criteria
- Level of performance
- Descriptors

Types of formative assessment

- Observations during in-class activities; of student‘s non-verbal feedback during lecture.
- Homework exercises as review for exams and class discussions)
- Reflections journals that are reviewed periodically during the semester
- Question and answer sessions, both formal (planned) and informal (spontaneous)
- Conferences between the instructor and student at various points in the semester
- In-class activities where students informally present their results
- Student feedback collected by periodically answering specific question about the instruction and their self-evaluation of performance and progress

Guidelines for essay type items

Following guidelines are for writing essay type items when developing a test.

1. Frame questions so that the examinee‘s task is explicitly defined.
2. Specify the value and an approximate time limit for each question.
3. Do not employ optional questions.
4. Employ a larger number of questions that require relatively short answers rather than only a few questions that require long answers.
5. Verify a question‘s quality by writing a trial response to the question.
6. Prepare a tentative scoring key in advance of considering examinee responses.
7. Score all answers to one question before scoring the next question.
8. Make prior decisions regarding treatment of such factors as spelling and punctuation.
9. Evaluate essay responses anonymously

Written test

In classroom assessment different forms of assessments are utilized. Each form of test has its own benefits and disadvantages. Most common type of assessment used in classrooms is written assessment

Types of Written Tests

- Verbal
- Non-verbal
- Objective

- Subjective
- Teacher Made
- Standardized
- Power
- Speed

Analytical rubric

Each criterion is assessed separately by using different descriptive ratings. Each criterion is given a separate score.

Final score is made up of adding each component parts. It takes more time to score but gives detailed more feedback.

Analytic scoring requires a list of major elements that students are expected to include in an ideal answer of a question.

Analytic rubric is used to score an essay type of question on different points and significantly it works best on the restricted response type essay question.

Advantages

- More detailed feedback.
- Scoring more consistent across students and grades.

Disadvantages

Time consuming to score.

Norm referenced and criterion reference

NRT

Type of test which tells us where a student stands compared to other students. It helps determining a student‘s place or rank among a group of similar students. Such kind of test is called norm-referenced test

Dimensions

- It provides estimate of ability in a variety of skills in much shorter time.
- NRT tend to be general. It measures variety of skills at same time but fails to measure them thoroughly.
- It‘s hard to make decisions regarding the mastery of student‘s skill in subject

Criterion Referenced Assessment

A second type of test tells us about student‘s level of proficiency in or mastery of some skill or set of skills. This is achieved by comparing a student‘s performance to a standard mastery called a criterion. Test that yields such information is called Criterion Referenced Test

Dimensions

- CRT tends to be specific. It measures particular set of skill at one time and focus on level of achievement of that skill. CRT gives clear picture regarding the mastery of student‘s skill in subject.
- It measures skill more thoroughly so naturally it takes more time comparing to NRT in measuring the mastery of said skill
- Items included in CRT are relatively easier. Around 80% of the students are expected to respond item correctly in the test
- CRT compares students ‘performance to standards indicative of mastery.
- Breadth of content sampled is narrow and covers very few objective

Reliability

Reliability refers to the consistency of measurement

- Reliability refers to the results obtained with an assessment instrument and not to the instrument itself.
- An estimate of reliability always refers to particular type of consistency (stability, equivalence, internal consistency)
- Reliability is necessary but not sufficient condition for validity.
- Reliability is primarily statistical (range +1 and -1).

Methods of reliability

- Test-Retest (stability)
- Equivalent Forms (equivalence)
- Test- Retest with Equivalent Forms (stability and equivalence)
- Split Half (Internal Consistency)
- Kuder- Richardson(Internal Consistency)
- Cronbach Alpha (Internal Consistency)
- Inter-rater Reliability (consistency of rating)

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Nature of validity-----5

Following are different points that describe the nature of validity.

1. Appropriateness of the interpretation of the results

Validity is referred as validity of test‖ but it is in fact validity of the interpretation and use to be made of the results.

2. Validity is matter of degree

It does not exist on all or none basis. It is best considered in term of categories that specify degree, such as high, moderate or low validity

3. Validity is specific to some particular use or interpretation

No assessment is valid for all purposes. An arithmetic test may have high degree of validity for computational skill and low degree for arithmetical reasoning

4. Validity is a unitary concept

Validity does not have different types. It is viewed as a unitary concept based on different kind of evidences

5. Validity involves an overall evaluative judgment

Elements of Rubric

A rubric includes:

1. Score

2. Criteria

3. Level of performance

4. Descriptors

Score

A system of number or values assigned to a work often combines with a level of performance. High numbers are for best performance like 4, 5 or 6 whereas down to 1 or 0 are the lowest score in a performance assessment.

Criteria

It tells us that which feature, trait or dimension to be measured and include a definition and example to make clear the meaning of each trait to be assessed.

Level of Performance

There are adjectives to describe the performance levels. These levels tell students what they are expected to do.

Descriptors can be used with levels of performance to achieve objectivity but they can be used without them as well.

Descriptors

These are details for each level of performance to reliable and unbiased scoring.

Formative Assessment

Formative Assessment determines learning progress, provides feedback to reinforce learning, and correct learning errors. When we assess student during classroom instruction with a purpose to have a feedback that how can we make our teacher learning process better, that is formative assessment. In this assessment, we are not assessing what students learnt or not rather we assess the process behind the students learning .The process behind the student learning includes teaching method , book, . If we make all these things according to the needs of students then learning will improve.

Summative Assessment----5

Summative assessment comes at the end of the instructional session (course of unit). It is designed to measure the extend of achievement of intended learning outcomes. The primary utility of this type of assessment is to assign grades and certifying the level of mastery and expertise in certain subject. It is usually done through teacher made achievement tests or alternative assessment techniques like portfolio to summarize the overall performance of the student at the end of the session. It is not compulsory to done it at the end of the semester, in

semester system there is midterm and final term in one semester, and these both are summative assessment.

Sociometric technique

Sociometric technique is method for assessing the social acceptance of individual students and the social structure group. It is based on students choice of companion for some group situation. This form was used to measure student‘s acceptance as seating companions, work companions and play companions.

There are few important principles of sociometric choosing

1. The choices should be real choices that are the natural part of classroom activities.

2. The basis for the choice and restriction on the choosing should be made clear.

3. All students should be equally free to participate in the activity or situation.

4. The choice of each student make must be kept confidential.

5. The choices should actually be used to organize or rearrange the group

Name of Psychomotor Domain

i. Perception

ii. Set

iii. Guided Response

iv. Mechanism

v. Complex covert Response

vi. Adaption

vii. Origination

Portfolio

Systematic collection of students work into portfolios can serve a variety of instructional and assessment purposes. The value of portfolios depend heavily on the clarity of purpose the guidelines for the inclusion of materials, and the criteria to be used in evaluating portfolio.

Portfolio of students work

A portfolio is collection of student work selected to serve a particular purpose such as documentation of student growth. It is purposeful; collection of pieces of student‘s work.

Key steps in defining and using portfolios

1. Specify purpose.

2. Provide guidelines for selecting portfolios.

3. Define student‘s role in selection and self-evaluation.

4. Specify evaluation criteria.

5. Use portfolios in instruction and communication.

Strengths of portfolios

1. The can be readily integrated with the instruction.

2. Provide opportunity to student‘s to show what they can do.

3. Encourage to become reflective learner.

Evaluation

Evaluation is process of making a value judgment against intended learning outcomes and behavior, to decide quality and extent of learning. Evaluation is always related to your purpose, you aligned your purpose of teaching with what students achieved at the end, with their quality and quantity of learning

Theory (IRT) and Item Analysis

Item Response Theory‖ (IRT) presents a model for expressing the association between an individual's response to an item and the underlying latent variable (often called "ability" or "trait")
The latent variable, expressed as theta (θ), is a continuous one-dimensional construct that explains the covariance among item responses
People at higher levels of latent trait have a higher probability of responding correctly or endorsing an item.

High stake testing

The use of test and assessment to make decisions that are of prominent educational, financial, or social impact.

Decision based on high stake testing

Promotion to next grade

Awarding of diploma or degree

Evaluation of school performance

Incentives and accountability of school staff

Holistic

Holistic rubric is probably more appropriate for extended response essay type items involving a student‘s abilities to synthesize and create and when no definite answer can be prespecified.

Bloom taxonomy

Bloom’s Taxonomy of Learning Objectives Bloom‘s Taxonomy was presented by Benjamin Bloom in the 1956, consists of a framework with most common objectives of classroom instruction.

Bloom’s Taxonomy of Learning Objectives

Those dealing in three different domains and further sub categories in these domains.

Cognitive

Affective

Psychomotor

Cognitive Domain:

Knowledge

Comprehension

Application

Analysis

Synthesis

Evaluation

Affective Domain

Receiving

Responding

Valuing

Organization

Characterization

Psychomotor Domain

Perception

Set

Guided Response

Mechanism

Complex covert Response

Adaption

Origination

Classical Test Theory (CTT) and Item Analysis

Classical Test Theory (CTT) has relatively weak theoretical assumptions, which make it easy to apply in many testing situations. Relatively weak theoretical assumptions not only characterize CTT but also its extensions (e.g., generalizability theory).
CTT‘s major focus is on test-level information, item statistics (i.e., item difficulty and item discrimination) are also an important part of the CTT model.

High Stake Testing-----5

Why does it matter?

The use of test and assessment to make decisions that are of prominent educational, financial, or social impact.

Decision based on high stake testing

1. Promotion to next grade

2. Awarding of diploma or degree

3. Evaluation of school performance

4. Incentives and accountability of school staff

High Stake Testing in Pakistan

In school education in Pakistan, high stake testing is conducted at each level from primary to higher secondary.

1. Grade 5 and 8 examination by Punjab Examination Commission.

2. Grade 10 and 12 examination by BISEs

Test construction in High Stake testing

1. The format of items are same as used in CRT in classroom

2. Item of developed by professional employed in dedicated organization

Item banks are developed

4. Psychometric properties of item are tested

5. The process goes on round the year

Criticism on high stake testing

High stake tests have same issues as classroom test but much larger impact and consequences.

Example of knowledge comprehension and application level

Knowledge:

It is defined as the remembering of previously learned material. This may involve the recall of a wide range of facts, procedures principals and generals, the recall of procedures and the processes.

Sample Question: Define the 6 levels of Bloom's taxonomy of the cognitive domain.

Comprehension:

It is defined as the ability to grasp the meaning of the material. Individual can make use of the content or idea being communicated without necessarily related it to other content and seeing its fullest implications.

Sample Question: explain the purpose of Bloom's taxonomy of the cognitive domain.

Application:

It refers to the ability to use the previously learned material in new and concrete situations. The abstractions may be in the shape of universal ideas, rules of methods.

Sample Question: write an instructional objective for each level of Bloom's taxonomy.

Item characteristic curve

To analyze items using IRT, the main thing need to consider is item characteristic curve (ICC). The item characteristic curve is considered as the basic building block of item response theory.

It gives a picture of:

The item difficulty

Discrimination power

The probability of answering correctly by guessing

Advantages of anecdotal record:

Following are the advantages of anecdotal records.

It depicts actual behaviors in natural situations.

Facilitate gathering evidence on the events that are exceptional but significant.

Beneficial for students with less communication skills.

Steps to design rubric

Basic Steps to design Rubric

Identify a learning goal.

Choose outcomes that may be measured.

Develop or adapt existing rubric.

Share it with students

Test security

Educational decisions

In the classroom, decisions are constantly being made. Teachers face huge numbers of dilemmas every day. These decisions can be of following nature

Types of Educational Decisions

- Instructional
- Grading
- Diagnostic
- Selecting
- Placement
- Counseling and Guidance
- Program or Curriculum
- Administrative

These types of decisions are taken at different levels. Some are decided at Board/Administrative level while some are taken at school management level and other are taken in classrooms by teachers

Scoring Rubrics for Essay Type Items

Scoring of essay items is a time consuming and difficult process. Reliability of the test demands that scoring should be consistent not only by the rater at different times but by two independent raters as well.

Scoring rubrics are descriptive scoring schemes that are developed by teachers or other evaluators to guide the analysis of writing products or processes of student‘s piece of writing or any performance

Alternatives

The list of suggested solutions which may include words, numbers, symbols or phrases are called alternatives (also called choices or options).

The correct alternative in each item is called the answer while all incorrect or less appropriate alternatives are called distractors or foiled and the student‘s task is to select the correct or best alternative from all the options.

Formative assessment

Formative assessment provides feedback and information during the instructional process, while learning is taking place, and while learning is occurring. Formative assessment measures student progress but it can also assess your own progress as an instructor.

Elements of rubric

A rubric includes:

Score

Criteria

Level of performance

Descriptors

Types of formative assessment

- Observations during in-class activities; of student‘s non-verbal feedback during lecture.
- Homework exercises as review for exams and class discussions)
- Reflections journals that are reviewed periodically during the semester
- Question and answer sessions, both formal (planned) and informal (spontaneous)
- Conferences between the instructor and student at various points in the semester
- In-class activities where students informally present their results
- Student feedback collected by periodically answering specific question about the instruction and their self-evaluation of performance and progress

Guidelines for essay type items

- Following guidelines are for writing essay type items when developing a test.
- Frame questions so that the examinee‘s task is explicitly defined.
- Specify the value and an approximate time limit for each question.

- Do not employ optional questions.
- Employ a larger number of questions that require relatively short answers rather than only a few questions that require long answers.
- Verify a question‘s quality by writing a trial response to the question.
- Prepare a tentative scoring key in advance of considering examinee responses.
- Score all answers to one question before scoring the next question.
- Make prior decisions regarding treatment of such factors as spelling and punctuation.
- Evaluate essay responses anonymously

Written test

In classroom assessment different forms of assessments are utilized. Each form of test has its own benefits and disadvantages. Most common type of assessment used in classrooms is written assessment

Types of Written Tests

- Verbal
- Non-verbal
- Objective
- Subjective
- Teacher Made
- Standardized
- Power
- Speed

Analytical rubric

Each criterion is assessed separately by using different descriptive ratings. Each criterion is given a separate score.

Final score is made up of adding each component parts. It takes more time to score but gives detailed more feedback.

Analytic scoring requires a list of major elements that students are expected to include in an ideal answer of a question.

Analytic rubric is used to score an essay type of question on different points and significantly it works best on the restricted response type essay question.

Advantages

More detailed feedback.

Scoring more consistent across students and grades.

Disadvantages

Time consuming to score.

Norm referenced and criterion reference

NRT

Type of test which tells us where a student stands compared to other students. It helps determining a student‘s place or rank among a group of similar students. Such kind of test is called norm-referenced test

Dimensions

It provides estimate of ability in a variety of skills in much shorter time.

NRT tend to be general. It measures variety of skills at same time but fails to measure them thoroughly.

It‘s hard to make decisions regarding the mastery of student‘s skill in subject

Criterion Referenced Assessment

A second type of test tells us about student‘s level of proficiency in or mastery of some skill or set of skills. This is achieved by comparing a student‘s performance to a standard mastery called a criterion. Test that yields such information is called Criterion Referenced Test

Dimensions

CRT tends to be specific. It measures particular set of skill at one time and focus on level of achievement of that skill. CRT gives clear picture regarding the mastery of student‘s skill in subject.

It measures skill more thoroughly so naturally it takes more time comparing to NRT in measuring the mastery of said skill

Items included in CRT are relatively easier. Around 80% of the students are expected to respond item correctly in the test

CRT compares students ‘performance to standards indicative of mastery.

Breadth of content sampled is narrow and covers very few objective

Reliability

Reliability refers to the consistency of measurement

Reliability refers to the results obtained with an assessment instrument and not to the instrument itself.

An estimate of reliability always refers to particular type of consistency (stability, equivalence, internal consistency)

Reliability is necessary but not sufficient condition for validity.

Reliability is primarily statistical (range +1 and -1).

Methods of reliability

Test-Retest (stability)

Equivalent Forms (equivalence)

Test- Retest with Equivalent Forms (stability and equivalence)

Split Half (Internal Consistency)

Kuder- Richardson(Internal Consistency)

Cronbach Alpha (Internal Consistency)

Inter-rater Reliability (consistency of rating)

Validity

Validity is an evaluation of adequacy and appropriateness of the interpretation and uses of results. It determines if a test is measuring what it intended to measure.

Reliability

Reliability refers to the consistency of assessment results.

Reliability vs Validity

Reliability of measurement is needed to obtain the valid results, but we can have reliability without validity. Reliability is necessity but not sufficient condition for validity

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2 marks

Q: what are subjective type questions? Tell names

ANS: Subjective test items (essay questions) are constructed response type questions that can be the best way to measure the students' higher order thinking skills, such as applying, organizing, synthesizing, integrating, evaluating, or projecting while at the same time providing a measure of writing skills. Essay items can vary from very lengthy (5-10 pages), open-ended to limited or restricted response (one page or less). Therefore, essay type questions are divided into two types

I. Restricted Response Items
II. Extended Response Items

Q: High stake testing matters

ANS: High-stakes testing as the use of a summative test or an assessment (i.e., a test designed to measure student achievement after a period of instruction has been completed) to make decisions that are of prominent educational, financial, or social impact.

Types of High stake test

Criterian Referenced Test

Norm-Referenced High-Stake Tests

Q: How to design rubric?

ANS: 1. Identify a learning goal.

2. Choose outcomes that may be measured.

3. Develop or adapt an existing rubric.

4. Share it with students

Q: Elements of Rubric

ANS: A rubric includes:

1. Score

2. Criteria

3. Levels of performance

4. Descriptors

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Q: Restricted response items names

ANS: Restricted response item: Pose a specific problem for which student needs to recall suitable information, organize it, derive a defensible conclusion, and express it within the given limits the questions (like 400 word, 30 min. etc.)

Some of which are;

1. Analysis of relationship.

2. Compare and contrast positions.

3. Explain cause-effect relationship.

4. Organize data and support a viewpoint.

5. Formulate hypotheses.

6. Point out strengths and weaknesses.

7. Integrate data from various resources

3 marks

Q: Advantage if essay type questions

Ans: Extended response questions have following advantages over other types of question formats: 1. Effective for assessing higher order abilities: analyses synthesize and evaluate. 2. It is comparatively less time consuming to develop such items. 3. Emphasizes essential communicational skills. 4. Guessing is eliminated

Q: State any three limitations of mcqs

Ans: Limitations of MCQs. 1. Take a long time to construct in order to avoid arbitrary and ambiguous questions. 2. Also require careful preparation to avoid preponderance of questions testimony recall. 3. Provide cues that do not exist in practice.

Q: analytic rubric? In which conditions it is used?

ANS: Analytic rubric

Each criterion is assessed separately by using different descriptive ratings. Each criterion is given a separate score. Final score is made up of adding each component parts. It takes more time to score but gives detailed more feedback.

When to use:

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1. Several faculty are collectively assessing student work.

2. Outside audience will be examining rubric scores.

3. Profiles of specific strength/weakness are desired.

Q: Benefits of rubric in evaluation process

ANS: A rubric is an explicit set of criteria used for assessing a particular type of work or performance and provides more details than a single grade or mark. It is a set of scoring guidelines for evaluating student work.

Q: Item analysis

ANS: The item analysis is an important phase in the development of a test. In this phase statistical methods are used to identify any test items that are not working well. If an item is too easy, too difficult, failing to show a difference between skilled and unskilled examinees, or even scored incorrectly, an item analysis will reveal it.

5 marks:

Q: How self assessment should be implemented

ANS: Boud (1995) argues that the way in which self-assessment is implemented is critical to its acceptance by students. According to Boud (1995), the implementation process needs to include:

A clear rationale: what are the purposes of this particular activity?

Explicit procedures—students need to know what is expected of them.

Reassurance of a safe environment in which they can be honest about their own performance without the fear that they will expose information which can be used against them.

Confidence that other students will do likewise, and that cheating or collusion will be detected and discouraged (Boud, 1995, p.182).

Q: Extended response type questions with example

ANS: It allows the students to determine the length and complexity of the response. This type of questions is most suitable for measuring higher level mental processes skills like synthesis and evaluation. Teachers prefer these questions when they want to have information about student's ability to organize, integrate, express and evaluate information.

Example: What are the teachings of Islam about the respect of elders? How do these teachings help us to make a good society? Marks will be given for correct information, your own point of view, good judgment and good presentation.

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Q: Restricted response type questions with example

ANS: Pose a specific problem for which student needs to recall suitable information, organize it, derive a defensible conclusion, and express it within the given limits of the questions (like 400 word, 30 min. etc.).

Example: List the similarities and differences in the process of cell division in meiosis and mitosis? A variety of learning outcomes (Linn, 2000) can be checked by using this format of essay question. Some of which are;

1. Analysis of relationship.

2. Compare and contrast positions.

3. Explain cause-effect relationship.

4. Organize data and support a viewpoint.

5. Formulate hypotheses.

6. Point out strengths and weaknesses.

7. Integrate data from various resources.

Q: What is report? Name of reports?

ANS: Reporting is the process of communicating the knowledge and understanding gained from assessing a student's work to improve learning

There are two forms of reports 1. Formal reports 2. Informal reports

Q: Explain formal and informal report

ANS: Formal reports

Formal reports communicate about important aspects of the students' progress in the areas of academic, social, human and career development to parents and students.

Informal reports

Informal report tells parents about what their child is able to do. It provides information about the areas of learning that need further attention or development and the ways the teacher is supporting the student's learning needs.

Q: assembling or packaging the test

ANS: Assembling Test Once you have measureable instructional objectives, test blueprint, written test items matching instructional objectives, so you are ready to

• package the test and • reproduce the test

Assembling Test (Packaging the Test)

1. Grouping together items of similar format

2. Arranging test items from easy to hard

3. Properly spacing items

4. Keeping items and options on the same page

5. Placing illustrations near the descriptions

6. Checking the answer key

Q: What is derived score? Tell name of any two types of derived score.

ANS: A derived score is a numerical report of test performance on a score scale that has well-defined characteristics and yields normative meaning. Most common type of derived scores are
1. Grade norms

2. Percentile norms

3. Standard norms

Q: What are subjective type questions? Also tell types.

ANS: Subjective test items (essay questions) are constructed response type questions that can be the best way to measure the students' higher order thinking skills, such as applying, organizing, synthesizing, integrating, evaluating, or projecting while at the same time providing a measure of writing skills. Essay items can vary from very lengthy (5-10 pages), open-ended to limited or restricted response (one page or less). Therefore, essay type questions are divided into two types
I. Restricted Response Items

II. Extended Response Items

Q: Develop 2 restricted response questions.

ANS: Q: List the similarities and differences in the process of cell division in meiosis and mitosis?

Q: Develop 2 extended response questions.

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ANS: Q1. Give examples to identify different forms of governments, their advantages and disadvantages over each other. Which form of government is most suitable in the socioeconomic and cultural context of our country? Your response will be graded on accuracy of information, power of arguments, conciseness and orderliness of presentation. It is suggested to keep your response limited to 5000-6000 words.

Q2. What are the teachings of Islam about the respect of elders? How do these teachings help us to make a good society?

Q: How to design a rubric.

ANS: Basic Steps to Design Rubric

1. Identify a learning goal.

2. Choose outcomes that may be measured.

3. Develop or adapt an existing rubric.

4. Share it with students.

Q: How to conduct parent teacher meeting.

ANS: 1. Make plans for the meeting.

2. Begin the meeting in a positive manner.

3. Present the student's strong points before describing the areas needing improvement.

4. Encourage parents to participate and share information.

5. Plan a course of action cooperatively.

6. End the meeting with a positive comment.

7. Use good human relation skills during the meeting

Q: Fixed choice assessment and complex performance assessment.

ANS: By format of assessment

i. Fixed Choice Assessment
ii. Complex Performance Assessment

Fixed Choice Assessment

Fixed Choice Assessment is sued to measure the skills of people efficiently (means measure more skills in less time) and for this we usually use fixed choice items i.e. Multiple choice question, matching exercise, fill in the blanks and true false. It is called fixed choice because the person who is attempting the paper does not need to write the answer, just need to choice the answer.

Complex Performance Assessment

Complex Performance assessment is used for measurement of performance in contexts and the problems valued in their own right. This includes hands on laboratory experiments, projects, essays, oral presentations.

Q: State 5 rules for constructing alternative form questions.

ANS: 1. Avoid broad general statements if they are to be judged true or false.

2. Avoid trivial statements

3. Avoid the use of negative especially double negative statements.

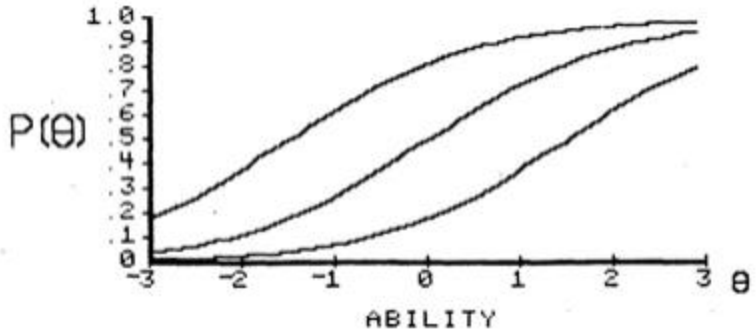
4. Avoid to include two ideas in one statement unless cause and effect relationship are being measured

5. Avoid long complex sentences

Q: What is Item difficulty in IRT? Explain with graph.

ANS: The application of Item difficulty in IRT is defined as the ability at which the probability of success on the item is .5 on a logit scale, which is also known as threshold difficulty (see Fig. 3). An item's level of difficulty is another factor affecting an individual's probability of responding in a particular way. An item that has a high level of difficulty will be less likely to be answered correctly by an examinee with low ability than an item that has a low level of difficulty (i.e., an easy item).

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REMEMBER ME IN PRAYERS